

Two-Timescale Resilience

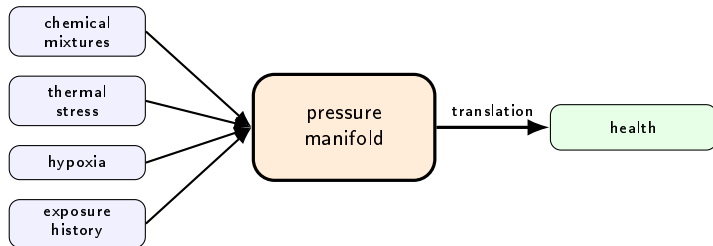
Mapping the Contraction of Biological Viability Bounds

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The Pressure Manifold

How do we construct a pressure manifold: a many-input device that funnels qualitatively different burdens *into* an account of health?



without resorting to *regulatory* limits as the *stand-in* for “health”.

The Regulatory Trap

What is Health?

In standard ecotoxicology, health is posited as a negative: **the absence of a threshold exceedance.**

"The organism is treated as a rigid fortress defending a baseline. This is not health; it is the description of a pathology."

Canguilhem: Health as Normative Capacity



Health is normativity.

True biological health is the positive, creative capacity to tolerate environmental shocks, absorb an engulfing medium, and establish a **new biological norm**.

"A state . . . which the living being's every anxious effort will tend to preserve from every eventual disturbance."

Energy as the Currency of Capacity

The Thermodynamic Assumption

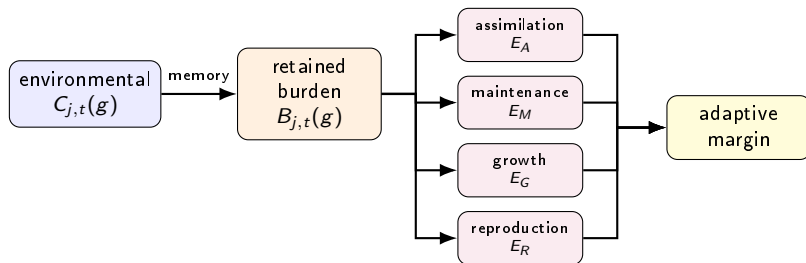
A norm transformation is accessible only if the biological system has enough **energetic luxury** to pay for it.

The Mechanistic Translation

How do we go from chemical concentrations to a quantifiable loss of norm-generating capacity?

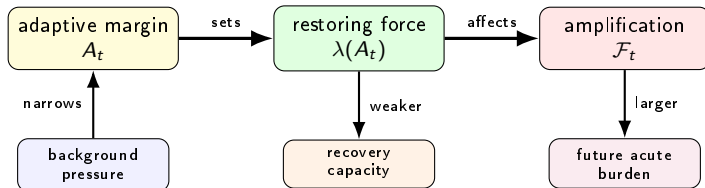
Dynamic Energy Budget (DEB) Engine

We map the phenomenological pressure of the environment into DEB thermodynamic taxes to estimate remaining **Adaptive Margin**.



Adaptive Margin is the remaining energetic distance between what the organism can mobilize, and what it **must** spend to stay alive.

From Adaptive Margin to Amplification

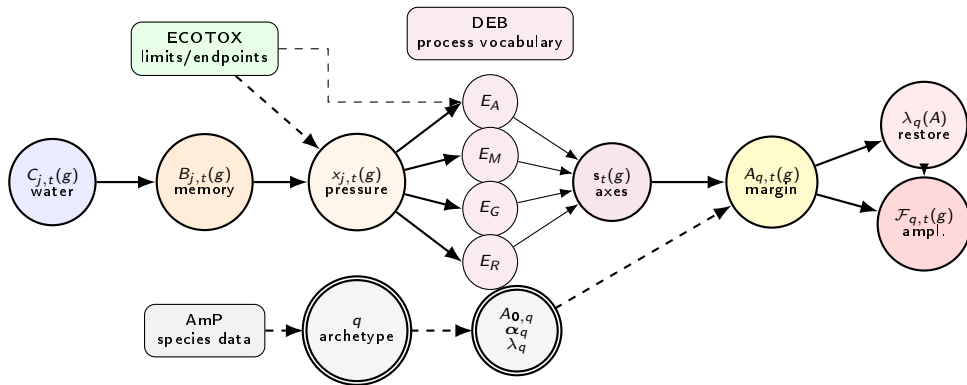


$$\lambda(A_t) = \lambda_{\min} + (\lambda_{\max} - \lambda_{\min}) \frac{\max(A_t, 0)}{K_A + \max(A_t, 0)}$$

$$\mathcal{F}_t = \frac{\lambda(A_0)}{\lambda(A_t)}$$

Lower adaptive margin means weaker recovery; weaker recovery means a larger burden from the same acute event.

The Epistemic Architecture

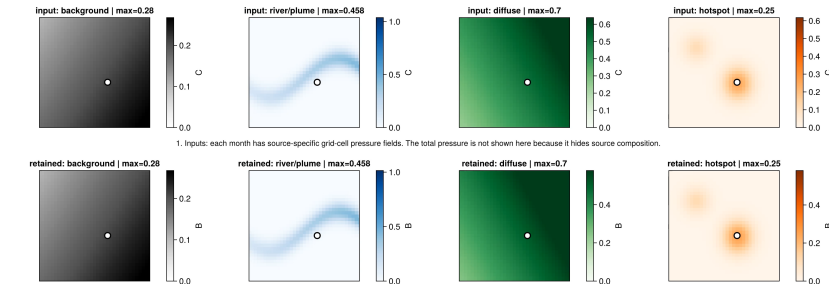


- **The Commitment:** We collapse DEB state-space matrices into a single macroscopic scalar (A_t).
- **The Trade-off:** We trade microscopic precision for spatial tractability. We model the *envelope of viability*, not exact life history.

The Theoretical Envelope in Space

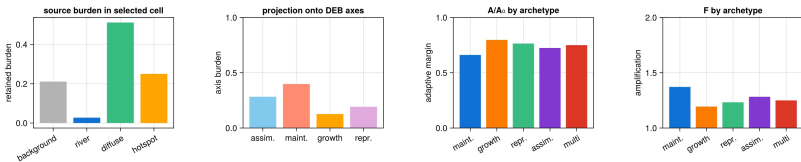
Spatial distribution of environmental pressure and resulting capacity.

Animated source-resolved story view — month 1, cell (22, 15)



1. Inputs: each month has source-specific grid-cell pressure fields. The total pressure is not shown here because it hides source composition.

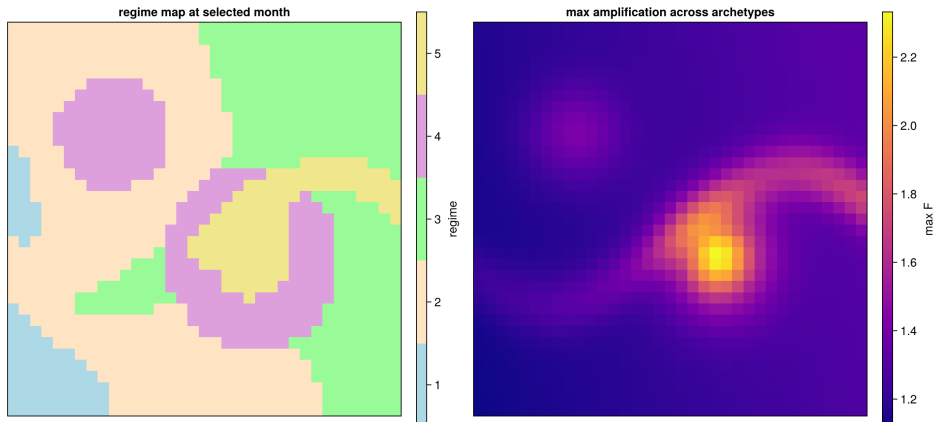
2. Memory: current pressure is carried forward as retained burden. Pulses can persist after the input field has weakened.



Embodied Memory and Biological Scar Tissue

Even when external concentration clears, retained internal burden keeps the Adaptive Margin compressed, leaving recovery capacity structurally compromised.

Time-varying synthetic regimes — month 26, $k = 5$



Closing

By fusing DEB theory with AmP data and ECOTOX pressure, we shift the fundamental question.

We stop asking: *"Is the water toxic?"*

And start asking: *"How much adaptive margin remains?"*

References I

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